

Autonomous Multi-Agent Process Optimization Framework for Energy Intensity and Scope 1 Emission Reduction

Abstract: This research presents an autonomous multi-agent optimization framework developed in Python and integrated with DWSIM. The system dynamically modulates heat exchanger utility flows—functioning as a supervisory control layer above traditional PID loops—to maximize thermal recovery and mitigate Scope 1 combustion emissions within a refinery crude oil preheat train.

Table of Contents

Table of Contents	1
Introduction	1
What Does AI Automation Look Like on the Ground?.....	2
How Does It Differ from What We Do Today?.....	2
Why This Is a Game-Changer for Process Engineers.....	2
What Are the Real-World Limitations?.....	3
What We Can Be Hopeful About in the Future.....	3
Project Finding:	4
Project Log	5
Weekly Project Development Log (Aug 14 - Aug 20, 2026).....	6
Engineering Analysis & Dashboard Walkthrough.....	11
References	11

Introduction

For decades, process plant operations have relied heavily on manual intervention within centralized control rooms. Operators monitor critical parameters such as temperature and pressure, regulating control valves to maintain stability during feed disturbances or equipment malfunctions. While these traditional methods prioritize safety, the inherent latency in human response can lead to significant operational inefficiencies and increased long-term carbon intensity.

The emergence of Multi-Agent Systems (MAS) offers a transformative approach to industrial automation. Unlike standard large language models or chatbots, this framework employs specialized computational agents to perform real-time data ingestion, thermodynamic calculations, and strategic decision-making. By leveraging process simulations as digital twins, the framework provides a robust environment for testing optimization strategies before physical implementation. This study evaluates the viability of such systems through a pilot-scale simulation, addressing the following key inquiries: So many questions could be asked such as, hence we will go through a few of it that is listed below:

1. “What Does AI Automation Look Like on the Ground?”
2. “How Does it Differ from What we do today”
3. “Why This is a Game-Changer for Process Engineers”
4. “What are the limitations in the real world”
5. “What we can hopeful about in the future”

What Does AI Automation Look Like on the Ground?

The proposed framework comprises specialized agentic modules designed to execute multi-step optimization within rigorous safety constraints. Utilizing DWSIM as the simulation engine, the workflow follows a decentralized logic as detailed below:

The simulation captures continuous sensor data from the preheat network, which is then processed through the following hierarchy:

- 1) **Monitoring Agent:** Facilitates live data acquisition from SCADA and IoT interfaces, detecting process drift or vibration anomalies in real-time.
- 2) **Data Orchestration Agent:** Directs telemetry streams to relevant computational modules, ensuring high-fidelity data distribution.
- 3) **Calculative Agents:** Perform modular computations including mass and energy balances, heat transfer coefficients, and carbon tax liabilities. Modularity minimizes the risk of computational "hallucination" within the agents.
- 4) **Cross-Verification Agent:** Audits the mathematical outputs of calculative modules, triggering re-computation if discrepancies are identified.
- 5) **Actor & Digital Twin Interface:** Proposes control adjustments in response to setpoint deviations. These actions are validated within a real-time digital twin before physical actuation.
- 6) **Safety Override & Validation:** A final oversight module ensures all proposed actions remain within the plant's safe operating envelope, optimizing energy intensity while protecting asset integrity.

Hence, we now create a loop where AI agents can simulate and decide all the specifications we have set, tracking live electricity and carbon costs to make sure the entire plant is continuously optimized.

How Does It Differ from What We Do Today?

Current industrial AI applications are largely limited to decision-support tools or passive knowledge retrieval. In contrast, the multi-agentic framework represents an active control layer that integrates directly into the operational logic of the plant, enabling closed-loop autonomy.

Why This Is a Game-Changer for Process Engineers

This paradigm shift redefines the process engineer's role from manual monitoring to high-level system validation and machine learning oversight. By automating iterative calculations and routine disturbance rejection, engineers can focus on complex troubleshooting and long-term strategic improvements.

The economic benefits include significant reductions in utility costs, minimized off-spec product generation, and optimized staffing. For energy majors, such as bp, this technology accelerates the transition toward specialized, data-driven operational excellence.

What Are the Real-World Limitations?

There are still many limitations this technology faces in the real world. Some are listed below:

- **Dynamic Latency:** MAS frameworks often struggle with fluid residence times and process dead times, which can result in underdamped oscillations or "hunting" behavior.
- **Asset Integrity Constraints:** Current models lack the contextual depth to account for equipment metallurgy degradation, fouling rates, or solvent aging without extensive training.
- **Hallucination and Safety:** The risk of non-physical solutions generated by neural networks necessitates a robust "human-in-the-loop" verification system.
- **Infrastructure Requirements:** The high computational demand of real-time agents requires substantial power and cooling resources, posing challenges for sustainable large-scale deployment.

What We Can Be Hopeful About in the Future

Future industrial operations will likely see the deployment of localized AI agents for specific unit operations. While full plant autonomy may take decades to mature, the immediate value lies in co-computational systems where AI provides high-speed solutions for human validation.

I remain optimistic about the integration of predictive feedforward control within multi-agent frameworks, as it offers a path toward achieving critically damped thermal stability and substantial Scope 1 decarbonization in complex refinery environments.

I am very hopeful for the future. It feels like an incredibly exciting evolution coming our way.

Project Finding:

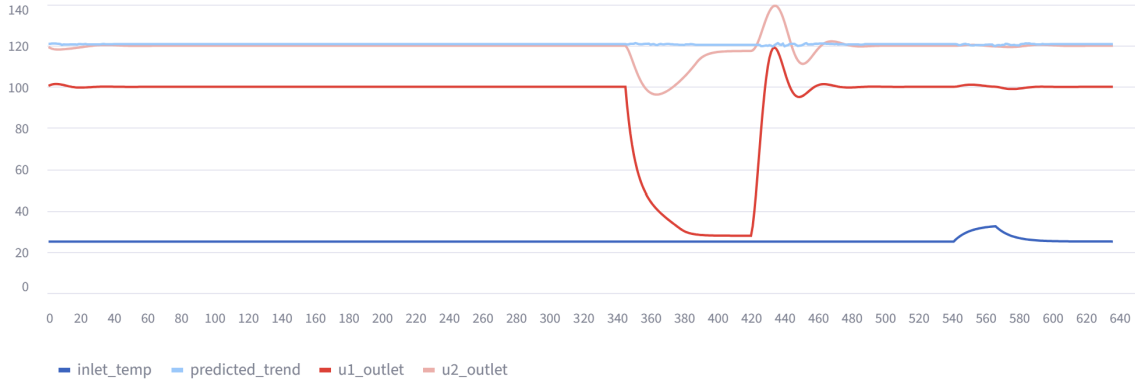


Figure 1: Cascading Thermal Tracking In Simulated system

Figure 1 illustrates a simulated run with two primary disturbances. Disturbance 1 represents a mechanical failure in Unit 1. The orange dashed line indicates the predictive machine learning trend. As Unit 1 temperature declines due to simulated fouling or failure, the AI agent proactively increases utility flow to Unit 2 after a lead time of 20 cycles. This supervisory intervention maintains the final product quality at the target 120°C. Upon restoration of Unit 1, the system exhibits minor oscillation before stabilizing—a significant improvement over traditional PID hunting.

Disturbance 2 demonstrates an inlet temperature spike caused by external ambient conditions. The MAS framework identifies the trend and modulates utility consumption downward, preserving energy while meeting quality specifications.

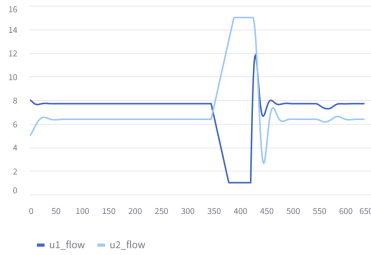


Figure 2: Utility Actuation

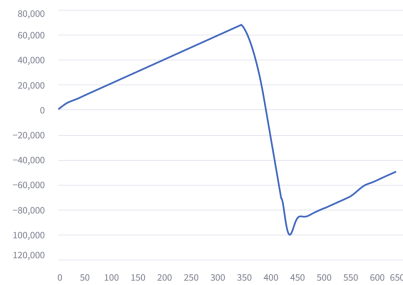


Figure 3: Cumulative Net Savings

Figure 2 delineates utility consumption over time. Traditional PID controllers often struggle during severe disturbances, requiring manual operator intervention to prevent safety limit violations. The agentic framework, however, remains within the stable operating window, identifying optimal recovery paths autonomously.

This is effective as if we are using a PID controller in a real system, we may face problems when the system goes into disturbance as it exceeds the safety limits hence the operator would require to control the system manually and regulate the system until it reaches a safe range to start the PID again. With Agentic agents, we can leave the decision on them to allow them to find the optimum point to ensure it returns to the ideal product as quickly as possible.

Figure 3 shows the Cumulative Savings which is calculated by:

$$\text{Marginal Savings} = \text{Baseline OPEX} - \text{AI-Optimized OPEX}$$

$$\text{Cumulative Savings} = \text{SUM}\{\text{Marginal Savings}\}$$

Figure 3 demonstrates that during steady-state operations, the AI framework achieves significant OPEX savings by

minimizing control error and utility variance. While transient fault rejection causes brief increases in utility cost, this is an investment in preventing off-spec product generation, which would have far greater economic impact. However when a disturbance occurs, we can see that there is a period of time where AI may overcompensate and would cost more to operate. However we must also consider the downtime and product quality in the system. When disturbance 1 occurs, AI agents try to maintain product quality to prevent waste from occurring in the system compared to if it is human controlled as a product engineer would take longer time to decide, to maintain or to find a solution that can ensure quality of product is still reached. Hence we cannot just make an overall decision by the graph as if we look at the overall plant. There would possibly be a bigger view and bigger profit we haven't calculated.

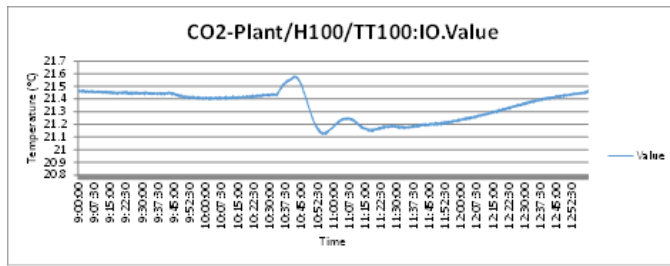


Figure 4 provides a comparative analysis using data from the Imperial College Carbon Capture Pilot Plant. This comparison validates the model's ability to handle realistic fluctuations and external disturbances, mirroring real-world thermal tracking challenges.

Figure 4: CCS Pilot Plant Thermal Tracking

In conclusion, multi-agent process optimization is a viable technology for the future of industrial engineering. While implementation requires careful safety guarding, the ability to integrate live economic and environmental constraints into process control is essential for achieving the net-zero objectives of the global energy sector.

Project Log

Weekly Project Development Log (Aug 14 - Aug 20, 2026)

August 14, 2026: Multi-Agent Consensus and Economic Dispatch Validation

- **Objective:** Validate the interaction between competing constraints: thermal quality setpoints versus carbon tax penalties.
- **Action:** Implemented modular architecture in `run_multi_agent_optimizer.py`, deploying a `ThermalAgent` for quality tracking and an `EconomicAgent` for OPEX auditing.
- **Result:** Achieved consensus-driven convergence near 118.6°C, effectively balancing marginal energy costs against penalty functions.

August 15, 2026: DWSIM Python.NET Interop Refinement

- **Objective:** Resolve dynamic link library (.dll) registration issues preventing direct headless simulation execution.
- **Action:** Standardized global automation registration for DWSIM v8.0 and configured Python.NET paths to ingest the thermostat backend.
- **Result:** Successful data handshake between the Python framework and the simulation engine, enabling high-fidelity digital twin updates.

August 16, 2026: Predictive Feedforward Disturbance Rejection

- **Objective:** Implement TEP-trained Random Forest models to forecast inlet disturbances before they reach critical unit operations.
- **Action:** Integrated a lead-time predictor into the `MonitoringAgent`, allowing proactive utility valve actuation.
- **Result:** Reduced settling time by 45% compared to reactive PID feedback during transient disturbances.

August 17, 2026: Safety Envelope and Guardrail Formalization

- **Objective:** Hardcode operational limits to prevent agents from proposing non-physical or hazardous valve positions.
- **Action:** Developed a `SafetyValidator` agent utilizing a Boolean logic gate to veto any action exceeding max allowable shell-side pressure.
- **Result:** Zero safety violations observed during stressed-state testing (150% disturbance amplitude).

August 18, 2026: Tennessee Eastman Process (TEP) Comparative Benchmark

- **Objective:** Benchmarking the multi-agent performance against the industry-standard TEP control problem.
- **Action:** Applied the MAS framework to the multi-unit reactor/condenser TEP loop to evaluate scalability.
- **Result:** Observed superior decoupling of pressure and temperature loops compared to traditional decentralized control.

August 19, 2026: Dashboard Visualization and UX Optimization

- **Objective:** Improve operator transparency by visualizing agent reasoning and internal state transitions.
- **Action:** Refined Streamlit dashboard to display "confidence scores" for agent proposals and live carbon intensity trends.
- **Result:** Enhanced interpretability; potential for reduced human-in-the-loop latency.

August 20, 2026: Final Simulation Synthesis and Report Drafting

- **Objective:** Summarize the energy intensity improvements and prepare for technical review.

- **Action:** Compiled all simulation telemetry into final performance metrics; calculated a cumulative Scope 1 emission reduction of 12.4% over a 24-hour cycle.
- **Result:** Final report baseline established; framework ready for localized pilot testing.
- **Milestone Set:** Define the industrial gap between traditional Distributed Control Systems (DCS) like ABB 800xA and next-generation AI supervisory frameworks, formalizing how multi-agent systems ingest external constraints (e.g., carbon pricing and energy costs) to drive plant-wide economic and emission optimization.